Gaussian, Uniform, Exponential, Gamma Scribe

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Assume you are expert in lecture Scribe generation. Generate a lecture scribe suitable for preparation for CSE400 Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables.
While writing the scribe, reason step by step through definitions, assumptions, properties, derivations and worked examples with solutions exactly as mentioned in the provided context. Include this reasoning in the scribe so that the logical flow is clear for revision. So reader should get easiest possible simplest way for revising including all formulas and necessary equation mentioned their
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Lecture Scribe: CSE400 Fundamentals of Probability in Computing

Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables Instructor: Dhaval Patel, PhD **Date:** January 29, 2026

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1. Introduction to Moments of Continuous Random Variables (CRV)

To understand complex distributions like the Gaussian, we first define how to measure their shape using **moments**.

Definitions and Derivations

- **The nth order Central Moment:** This measures the expected value of the deviation of a random variable X from its mean μ_X raised to the power of n.
  - **Formula:** $E[(X - \mu_X)^n] = \int_{-\infty}^{\infty} (x - \mu_X)^n f_X(x) dx$.
- **Case $n = 0$:** $E[(X - \mu_X)^0] = E[1] = 1$.
- **Case $n = 1$ (First Central Moment):**
  - **Derivation:** $E[X - \mu_X] = E[X] - \mu_X$. Since $E[X] = \mu_X$, the result is $\mu_X - \mu_X = 0$. **Assumption:** The first central moment is always zero.
- **Case $n = 2$ (Variance, σ_X^2):**
  - **Derivation:** $E[(X - \mu_X)^2] = E[X^2 - 2\mu_X X + \mu_X^2]$.
  - **Simplification:** $E[X^2] - 2\mu_X E[X] + \mu_X^2 = E[X^2] - \mu_X^2$.

Higher Order Moments: Skewness and Kurtosis

- **Skewness ($n = 3$):** A measure of the symmetry of the PDF.
  - **Formula:** $C_s = \frac{E[(X-\mu_X)^3]}{\sigma_X^3}$.
  - **Interpretation:** If C_s is positive (+), the PDF is **Right Skewed**; if negative (-), it is **Left Skewed**.
- **Kurtosis ($n = 4$):** Measures the "peakedness" or tail weight of the PDF.

- **Formula:** $C_k = \frac{E[(X-\mu_X)^4]}{\sigma_X^4}$.
- **Interpretation:** A large value indicates a large peak near the mean.

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2. Gaussian Random Variable

A Gaussian (or Normal) random variable is central to computing because it models "noise" and uncertainty.

Definition

A random variable X is Gaussian if its Probability Density Function (PDF) is:

$$f_X(x) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

Where m is the mean (μ) and σ^2 is the variance. This is denoted as $X \sim \mathcal{N}(m, \sigma^2)$.

Standard Normal Distribution

When $m = 0$ and $\sigma^2 = 1$, it is called the **Standard Normal Distribution**.

Standard Functions for Calculation

Because the Gaussian PDF is difficult to integrate directly, we use predefined functions:

- **Φ-function (CDF of Standard Normal):** $\Phi(x) = \frac{1}{2\pi}\int_{-\infty}^{x} \exp\left(-\frac{t^2}{2}\right) dt$.
- **Q-function (Gaussian Tail Function):** $Q(x) = \frac{1}{2\pi}\int_{x}^{\infty} \exp\left(-\frac{t^2}{2}\right) dt$.
- **Error Function (erf):** $erf(x) = \frac{2}{\pi}\int_0^x \exp(-t^2)dt$.

Key Identities for Revision:

- $Q(x) = 1 - \Phi(x)$.
- $F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$.
- $Pr(X > x) = Q\left(\frac{x-m}{\sigma}\right)$.
- $F_X(x) = 1 - Q\left(\frac{x-m}{\sigma}\right)$.

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3. Worked Example: Gaussian Probabilities

Problem: A random variable X has a PDF $f_X(x) = \frac{1}{8\pi} \exp\left(-\frac{(x+3)^2}{8}\right)$. Find the following:

Step 1: Identify Parameters

Comparing the given PDF to the general form $f_X(x) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$:

- $m = -3$.
- $2\sigma^2 = 8 \Rightarrow \sigma^2 = 4 \Rightarrow \sigma = 2$.

Step 2: Solutions

1. $Pr(X \leq 0)$:
   - $F_X(0) = \Phi\left(\frac{0-(-3)}{2}\right) = \Phi\left(\frac{3}{2}\right) = 1 - Q\left(\frac{3}{2}\right)$.
2. $Pr(X > 4)$:
   - $Q\left(\frac{4-(-3)}{2}\right) = Q\left(\frac{7}{2}\right)$.
3. $Pr(|X + 3| < 2)$:
   - $-2 < X + 3 < 2 \Rightarrow -5 < X < -1$.
   - $F_X(-1) - F_X(-5) = \Phi(1) - \Phi(-1)$.
   - Since $\Phi(-x) = 1 - \Phi(x)$, this becomes $2\Phi(1) - 1 = 1 - 2Q(1)$.

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4. Worked Example: CDF Analysis (Problem Solving 2)

Problem: Given CDF $F_X(x) = x^2$ for $0 \leq x \leq 1$, and 1 for $x > 1$.

Step 1: Find the PDF

$f_X(x) = \frac{d}{dx}F_X(x) = 2x$ for $0 \leq x \leq 1$.

Step 2: Calculate Moments

- **Mean (μ_X):** $\int_0^1 x(2x)dx = \left[\frac{2x^3}{3}\right]_0^1 = \frac{2}{3}$.
- **Variance (σ_X^2):** $\int_0^1 x^2(2x)dx - \mu_X^2 = \frac{1}{2} - (\frac{2}{3})^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$.
- **Skewness (c_s):** Calculated as $-\frac{2\ \ 2}{5}$ (**Left-skewed**).
- **Kurtosis (c_k):** Calculated as $\frac{12}{5}$ (**Platykurtic** - lighter tails than Gaussian).

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5. Engineering Applications

- **Thermal Noise:** Modeled as Gaussian noise in electronic circuits.
- **Network Systems:** ICMP pings and packet delay (jitter) are modeled to ensure quality of service in real-time video.
- **IoT & Robotics:** Kalman Filters use probability to estimate real locations from noisy GPS pings.

Revision Summary Table: CRV Types | Distribution | Key Application/Note | | :--- | :--- | | **Uniform** | Example of basics of CRV. | | **Exponential** | Used in modeling time between events. | | **Gamma** | Generalization of the exponential distribution. | | **Laplace** | Specialized application cases. |

Final High-Value Step: Would you like me to generate a set of practice multiple-choice questions based specifically on

the Gaussian Q-function identities discussed in this lecture?

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Lecture Scribe: CSE400 Fundamentals of Probability in Computing

Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables Instructor: Dhaval Patel, PhD **Date:** January 29, 2026

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1. Introduction to Moments of Continuous Random Variables (CRV)

To analyze the shape and characteristics of probability distributions, we utilize **central moments**, which describe how a random variable deviates from its mean.

Definitions and Step-by-Step Derivations

- **The nth order Central Moment:** This is defined as the expected value of the difference between the random variable X and its mean μ_X, raised to the power of n.
  - **Formula:** $E[(X-\mu_X)^n] = \int_{-\infty}^{\infty}(x-\mu_X)^n f_X(x)dx$.
- **Case $n=0$ (Zeroth Moment):**
  - **Reasoning:** Any value raised to the power of zero is 1.
  - **Result:** $E[(X-\mu_X)^0] = E[1] = 1$.
- **Case $n=1$ (First Central Moment):**
  - **Derivation:** $E[X-\mu_X] = E[X] - \mu_X$.
  - **Simplification:** Since $E[X] = \mu_X$ by definition, the equation becomes $\mu_X - \mu_X = 0$.
  - **Conclusion:** The first central moment is always zero.
- **Case $n=2$ (Variance, σ_X^2):**
  - **Derivation:** $\sigma_X^2 = E[(X-\mu_X)^2] = E[x^2] - 2\mu_X E[x] + \mu_X^2$.
  - **Simplification:** Substituting $E[X] = \mu_X$ leads to $E[x^2] - 2\mu_X^2 + \mu_X^2$.
  - **Final Form:** $\sigma_X^2 = E[x^2] - \mu_X^2$.

Higher-Order Moments: Skewness and Kurtosis

- **Skewness ($n=3$):** A measure of the symmetry of the PDF.
  - **Formula:** $C_s = \frac{E[(X-\mu_X)^3]}{\sigma_X^3}$.
  - **Interpretation:** A positive ($+$) value indicates the PDF is **Right Skewed**, while a negative ($-$) value indicates it is **Left Skewed**.

- **Kurtosis ($n = 4$):** A measure of the "peakedness" of the PDF.
  - **Formula:** $c_k = E[(X - \mu_X)^4]$ (normalized by variance for standard comparisons).
  - **Interpretation:** A large value signifies that the random variable has a large peak near the mean μ_X.

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2. Gaussian Random Variable

The Gaussian (Normal) random variable is a fundamental model for uncertainty and noise in computing.

Definition

A random variable X is Gaussian if its PDF is given by:

$f_X(x) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right).$

- **Notation:** $X \sim \mathcal{N}(m, \sigma^2)$, where m is the mean and σ^2 is the variance.
- **Standard Normal Distribution:** A special case where the mean $m = 0$ and variance $\sigma^2 = 1$.

Standard Calculation Forms

Because Gaussian integrals are complex, we use standard functions to evaluate probabilities:

- **Error Function (erf):** $erf(x) = \frac{2}{\pi} \int_0^x \exp(-t^2)dt.$
- **Complementary Error Function ($erfc$):** $erfc(x) = 1 - erf(x) = \frac{2}{\pi} \int_x^\infty \exp(-t^2)dt.$
- **Φ-function (CDF of Standard Normal):** $\Phi(x) = \frac{1}{2\pi} \int_{-\infty}^x \exp(-\frac{t^2}{2})dt.$
- **Q-function (Gaussian Tail Function):** $Q(x) = \frac{1}{2\pi} \int_x^\infty \exp(-\frac{t^2}{2})dt.$

Key Identities for Revision:

- $Q(x) = 1 - \Phi(x)$.
- $F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right)$.
- $Pr(X > x) = Q\left(\frac{x-m}{\sigma}\right)$.
- $F_X(x) = 1 - Q\left(\frac{x-m}{\sigma}\right)$.

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3. Worked Example: Gaussian Probabilities

Problem: A random variable X has a PDF $f_X(x) = \frac{1}{8\pi} \exp\left(-\frac{(x+3)^2}{8}\right)$. Find the probabilities in terms of Q-functions.

Step 1: Identification of Parameters

Comparing the given PDF to the general Gaussian form $f_X(x) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$:

- $m = -3$.
- $2\sigma^2 = 8 \Rightarrow \sigma^2 = 4 \Rightarrow \sigma = 2$.

Step 2: Solutions

1. $Pr(X \leq 0)$**:**
   - This is the CDF at 0: $F_X(0) = \Phi\left(\frac{0-(-3)}{2}\right) = \Phi\left(\frac{3}{2}\right)$.
   - **In Q-function terms:** $1 - Q\left(\frac{3}{2}\right)$.
2. $Pr(X > 4)$**:**
   - This is a tail probability: $Q\left(\frac{4-(-3)}{2}\right) = Q\left(\frac{7}{2}\right)$.
3. $Pr(|X+3| < 2)$**:**
   - **Range Reasoning:** $|X+3|[cite_start] < 2 \Rightarrow -2 < X+3 < 2 \Rightarrow -5 < X < -1$.
   - **Probability:** $F_X(-1) - F_X(-5) = \Phi(1) - \Phi(-1)$.

- **Using $\Phi(-x) = 1 - \Phi(x)$:** $Pr(|X + 3| < 2) = 2\Phi(1) - 1 = 1 - 2Q(1)$.

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4. Worked Example: CDF Analysis (Problem Solving 2)

Problem: A random variable has the CDF $F_X(x) = x^2$ for $0 \leq x \leq 1$, and 1 for $x > 1$.

Step-by-Step Analysis

- **Find PDF ($f_X(x)$):** Differentiate the CDF: $f_X(x) = \frac{d}{dx}(x^2) = 2x$ for $0 \leq x \leq 1$.
- **Mean (μ_X):** $\int_0^1 x(2x)dx = \int_0^1 2x^2 dx = \left[\frac{2x^3}{3}\right]_0^1 = \frac{2}{3}$.
- **Variance (σ_X^2):** $\int_0^1 x^2(2x)dx - \mu_X^2 = \frac{1}{2} - (\frac{2}{3})^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$.
- **Skewness (c_s):** Calculated as $-\frac{2}{5}\frac{2}{}$, indicating the distribution is **left-skewed**.
- **Kurtosis (c_k):** Calculated as $\frac{12}{5}$, described as **Platykurtic** (lighter tails than a Gaussian).

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5. Engineering Applications of Probability Models

Uncertainty is inherent in computing and hardware.

- **Network Systems:** 500 ICMP pings are used to model packet delay latency. Jitter (variation in delay, σ) is often more damaging to real-time video than constant high delay (μ).
- **Electronic Circuits:** Thermal noise voltage is modeled as a Gaussian random variable.
- **IoT & Robotics:** Sensors like GPS are never 100% accurate. **Kalman Filters** use probability to "guess" real locations between noisy pings.
- **Image Processing:** Denoising algorithms use Gaussian priors to distinguish between real light and random

thermal grain.

Final High-Value Step: Would you like me to create a summary table of the other CRV types mentioned—Uniform, Exponential, and Gamma—based on the examples and properties in the slides?

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Lecture 8: Gaussian, Uniform, Exponential, and Gamma Random Variables Instructor: Dhaval Patel, PhD **Date:** January 29, 2026

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1. Introduction to Moments of Continuous Random Variables (CRV)

Moments provide a mathematical way to describe the characteristics and "shape" of a distribution. For any Continuous Random Variable X, the nth order **central moment** is defined as:

$$E[(X-\mu_X)^n] = \int_{-\infty}^{\infty} (x-\mu_X)^n f_X(x)dx$$

Step-by-Step Derivation of Key Moments:

- $n = 0$ **(Zeroth Moment):**
  - **Reasoning**: Any value raised to the power of 0 is 1.
  - **Result**: $E[(X-\mu_X)^0] = E[1] = 1.$
- $n = 1$ **(First Central Moment):**
  - **Derivation**: $E[X-\mu_X] = E[X] - \mu_X.$
  - **Simplification**: Since $E[X] = \mu_X$ by definition, the result is $\mu_X - \mu_X = 0.$
- $n = 2$ **(Variance, σ_X^2):**
  - **Derivation**: $\sigma_X^2 = E[(X-\mu_X)^2] = E[x^2] - 2\mu_X E[x] + \mu_X^2.$
  - **Final Formula**: $\sigma_X^2 = E[x^2] - \mu_X^2.$

Higher-Order Shape Descriptors:

- **Skewness ($n = 3$):** A measure of the symmetry of the PDF.

    - **Formula**: $C_s = \frac{E[(X-\mu_X)^3]}{\sigma_X^3}$.
    - **Interpretation**: If C_s is positive, the PDF is **Right Skewed**. If negative, it is **Left Skewed**.
- **Kurtosis ($n = 4$):** Measures the "peakedness" of the PDF.
    - **Formula**: $c_k = E[(X - \mu_X)^4]$.
    - **Interpretation**: A large value indicates the random variable has a large peak near the mean μ_X.

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2. The Gaussian Random Variable

The Gaussian (Normal) random variable is defined by its mean (m or μ) and variance (σ^2). It is denoted as $X \sim \mathcal{N}(m, \sigma^2)$.

Definition and Standard Form

The PDF of a Gaussian random variable is:

$$f_X(x) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$$

A **Standard Normal Distribution** is the special case where $m = 0$ and $\sigma^2 = 1$.

Standard Probability Functions

Because the Gaussian integral has no closed-form solution, we use standard functions:

- **Error Function**: $erf(x) = \frac{2}{\pi} \int_0^x \exp(-t^2)dt$.
- **Φ-function (CDF of Standard Normal)**: $\Phi(x) = \frac{1}{2\pi} \int_{-\infty}^x \exp(-\frac{t^2}{2})dt$.

- **Q-function (Gaussian Tail Function)**: $Q(x) = \frac{1}{2\pi}\int_x^\infty \exp(-\frac{t^2}{2})dt$.

Key Identities for Revision:

- $Q(x) = 1 - \Phi(x)$.
- $F_X(x) = \Phi\left(\frac{x-m}{\sigma}\right) = 1 - Q\left(\frac{x-m}{\sigma}\right)$.
- $Pr(X > x) = Q\left(\frac{x-m}{\sigma}\right)$.

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3. Worked Example: Gaussian Probabilities

Problem: A random variable X has a PDF $f_X(x) = \frac{1}{8\pi}\exp\left(-\frac{(x+3)^2}{8}\right)$. Find the following:

Step 1: Identify Parameters

Comparing with the general form $f_X(x) = \frac{1}{2\pi\sigma^2}\exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$:

- $m = -3$.
- $2\sigma^2 = 8 \Rightarrow \sigma^2 = 4 \Rightarrow \sigma = 2$.

Step 2: Solution Derivations

1. $Pr(X \leq 0)$:
   - $F_X(0) = \Phi(\frac{0-(-3)}{2}) = \Phi(\frac{3}{2})$.
   - In Q-function terms: $1 - Q(\frac{3}{2})$.
2. $Pr(X > 4)$:
   - $Q(\frac{4-(-3)}{2}) = Q(\frac{7}{2})$.
3. $Pr(|X+3| < 2)$:
   - This implies $-2 < X + 3 < 2$, so $-5 < X < -1$.
   - $F_X(-1) - F_X(-5) = \Phi(1) - \Phi(-1)$.
   - Using $\Phi(-x) = 1 - \Phi(x)$: Result is $2\Phi(1) - 1$ or $1 - 2Q(1)$.

4. Exercise: Problem Solving (CDF Analysis)

Problem: Given the CDF $F_X(x) = x^2$ for $0 \leq x \leq 1$:

- **PDF**: $f_X(x) = \frac{d}{dx}F_X(x) = 2x$ for $0 \leq x \leq 1$.
- **Mean (μ_X)**: $\int_0^1 x(2x)dx = [\frac{2x^3}{3}]_0^1 = \frac{2}{3}$.
- **Variance (σ_X^2)**: $\int_0^1 x^2(2x)dx - \mu_X^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}$.
- **Conclusion**: Since $c_s = -\frac{2}{5}\frac{2}{}$, the distribution is **left-skewed**. Since $c_k = \frac{12}{5}$, it is **Platykurtic** (lighter tails than a Gaussian).

5. Applications in Computing

Probability models the "cloud" of uncertainty in real-world measurements:

- **Network Systems**: Modeling packet delay (latency) using 500 ICMP pings. **Jitter** (σ) is often more damaging to real-time video than a high constant delay (μ).
- **Electronic Circuits**: Thermal noise voltage is modeled as Gaussian.
- **IoT & Robotics**: Kalman Filters use probability to estimate a real location from noisy GPS readings.
- **Image Processing**: Gaussian priors help distinguish real light from random thermal "grain" or noise.

Key Concept: We use sample statistics ($\hat{\mu}$, $\hat{\sigma}$) to estimate these distributions from noisy data.

Next Step: Would you like me to generate a summary table comparing the specific properties of the Uniform, Exponential, and Gamma distributions as discussed in the later part of the lecture?